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Repeat collaboration and scientific innovation: evidence from dynamic ego networks of Nobel laureates

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This study explores how collaboration dynamics influence scientific innovation in Physics, Chemistry, and Physiology or Medicine by analyzing the evolving ego networks of Nobel laureates. Using data from the Microsoft Academic Graph and an updated Nobel laureate dataset (1900–2024), we quantify collaboration patterns through collaboration length and cumulative coauthorship strength within dynamic ego networks. Innovation is measured via four field- and time-normalized metrics: Citation, Novelty, Disruption, and Interdisciplinarity. By studying Nobel laureates, exemplars of transformative science, this work tests and extends theories of collaboration dynamics in elite scientific networks, offering insights into the mechanisms driving innovation. Our findings reveal that Nobel-winning papers predominantly stem from new collaborations rather than sustained partnerships. Regression analyses show a negative association between repeat collaboration and innovation, with prolonged coauthoring linked to reduced novelty, disruptiveness, and interdisciplinarity, though this effect is less pronounced in Chemistry, suggesting field-specific dynamics. Career age differences between coauthors mediate this relationship, with larger gaps amplifying negative impacts on citation and disruptiveness, except in Chemistry, where the trend diverges. These insights highlight the value of fresh collaborations in driving transformative science in these fields and suggest tailored strategies for fostering innovation.

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Introduction

Collaboration has become a cornerstone of modern science, particularly in fields like Physics, Chemistry, and Physiology or Medicine, where complex research questions demand interdisciplinary expertise and shared resources (Wuchty et al., 2007). However, the rise of collaborative research varies across disciplines, with international collaborations growing but not yet the norm in many fields (Nielsen and Andersen, 2021). Collaborative efforts consistently outperform solo-authored works in productivity and citation impact, underscoring the benefits of collective intellectual endeavor (Jones et al., 2008). Trust and effective communication within partnerships further amplify their potential to drive discovery (Fronczak et al., 2022; Li et al., 2013). Yet, while the advantages of collaboration are well-documented, the dynamics of repeat collaboration—sustained coauthoring over time—and its impact on scientific innovation in specific disciplines remain underexplored. Prolonged partnerships may enhance productivity through accumulated expertise and trust (Petersen, 2015), but they could also limit innovation by constraining exposure to diverse perspectives, as suggested by Granovetter's (1973) theory of weak ties.

Scientific innovation, encompassing originality, transformative potential, and intellectual breadth, extends beyond mere productivity (Fortunato et al., 2018; Uzzi et al., 2013; Wang et al., 2013). Traditional metrics like citation counts capture influence but fall short in assessing qualitative dimensions of innovation (Liu et al., 2023). Emerging measures—such as Novelty, which quantifies original knowledge recombination (Uzzi et al., 2013); Disruption, which assesses shifts in research trajectories (Funk and Owen-Smith, 2017); and Interdisciplinarity, which reflects disciplinary integration (Leydesdorff et al., 2019; Stirling, 2007)—provide a more comprehensive lens. Prior studies suggest that team composition and dynamics shape these innovation outcomes, with diverse and fluid collaborations often linked to greater originality (Lin et al., 2023; Palla et al., 2007; Wu et al., 2019; Xu et al., 2022a; Yang et al., 2024a). However, the role of repeat collaboration, characterized by its duration and frequency, in driving transformative outcomes remains ambiguous, particularly among elite scientists.

This study addresses this gap by examining repeat collaboration and scientific innovation in Physics, Chemistry, and Medicine through the careers of Nobel laureates, whose transformative contributions offer a unique lens to test theories of collaboration dynamics (Akbaritabar et al., 2024). As exemplars of scientific excellence, Nobel laureates operate within elite networks that disproportionately shape scientific impact, aligning with stratified structures observed in science (Akbaritabar et al., 2024; Nielsen and Andersen, 2021). Drawing on Granovetter's (1973) theory, we hypothesize that new collaborations (weak ties) foster novel and disruptive outcomes by facilitating diverse information flows, while repeat collaborations (strong ties) may lead to cognitive lock-in, reducing innovation (Johnson et al., 2003).

Our findings reveal that Nobel-winning papers in Physics, Chemistry, and Medicine predominantly stem from new collaborations rather than sustained partnerships, with repeat collaboration negatively associated with innovation, though this effect is weaker in Chemistry. Career age differences between coauthors mediate this relationship, with larger gaps amplifying negative impacts, except in Chemistry, where field-specific dynamics suggest a reliance on cumulative expertise (Kreimer, 2022). These insights advance Science of Science (SciSci) by quantifying collaboration's role in elite scientific networks and contribute to Science and Technology Studies (STS) by highlighting how globalized yet localized practices shape innovation (Kreimer, 2022). By focusing on these three fields, we uncover both universal and discipline-specific patterns, offering tailored strategies for fostering transformative research.

Background

Collaboration has become a defining feature of modern science, propelled by the growing complexity of research questions that demand interdisciplinary expertise and pooled resources (Wuchty et al., 2007). Collaborative papers consistently outpace solo-authored works in productivity and citation impact, underscoring the advantages of shared knowledge and collective effort (Jones et al., 2008). Trust and effective communication underpin these partnerships, creating fertile ground for discovery (Fronczak et al., 2022; Li et al., 2013; Xu et al., 2022b). However, while collaboration boosts output, its impact on innovation—particularly the generation of novel and disruptive ideas—remains contested. Evidence suggests that team dynamics and composition critically shape outcomes, with diverse, dynamic groups often linked to greater originality (Lin et al., 2023; Palla et al., 2007; Wu et al., 2019; Xu et al., 2022a; Yang et al., 2024a). Granovetter's (1973) theory of weak ties provides a theoretical foundation for this study, suggesting that new collaborations (weak ties) facilitate novel information flows, while repeat collaborations (strong ties) may entrench familiar approaches, potentially limiting innovation.

Scientific innovation is multifaceted, requiring robust, standardized metrics to capture its dimensions. Citation count, a long-standing proxy for influence, reflects a paper's reach within the academic community (Fortunato et al., 2018; Wang et al., 2013). Novelty, which measures the originality of knowledge recombination, builds on Uzzi et al. (2013), who quantify it through atypical journal pairings in citation networks—a method enhanced here with percentile rankings for cross-field comparability. Disruption, introduced by Funk and Owen-Smith (2017) and expanded by Wu et al. (2019), gauges a paper's capacity to shift research trajectories, though recent studies note a decline in such transformative outputs over time (Park et al., 2023). Interdisciplinarity, assessed via the Rao-Stirling index, captures the diversity of disciplinary influences, highlighting the breadth of intellectual integration (Leydesdorff et al., 2019; Stirling, 2007). Together, these metrics provide a comprehensive lens to evaluate innovation beyond mere productivity, addressing its qualitative impact on scientific progress.

Repeat collaboration, characterized by sustained coauthoring over time, stands apart from transient partnerships due to its duration and frequency (Petersen, 2015; Zeng et al., 2021). Such enduring ties are a cornerstone of scientific practice, fostering accumulated expertise and trust that drive productivity. Longitudinal analyses reveal that repeat collaborators often achieve higher citation counts, reflecting the stability of these relationships (Petersen, 2015). Yet, prolonged collaboration may carry hidden costs. Studies of patent innovation suggest that small, fluid teams outperform larger, entrenched ones in generating disruptive advances (Lin et al., 2023; Wu et al., 2019; Xu et al., 2022a). Similarly, in science, sustained partnerships risk entrenching familiar approaches, limiting exposure to fresh perspectives and potentially stifling novelty (Zeng et al., 2022).

Emerging research highlights broader systemic effects of repeat collaboration. Centralized collaboration networks may concentrate resources and attention, but they can also constrain diffusion of knowledge, forming socio-epistemic bubbles that presage declines in a field's relevance (Kang et al., 2025). This dynamic encourages conservative research strategies—exploring well-trodden paths rather than venturing into riskier, uncharted territory—potentially slowing progress, especially in mature disciplines (Rzhetsky et al., 2015; Sourati and Evans, 2023). Hill and Stein (2025) further estimate that such inefficiencies have imposed substantial costs, ranging from $1.5 to $8.8 billion since 1971, due to the lower quality of outputs from highly competitive, centralized structures. For junior

scientists, the rise of large, stable teams exacerbates challenges, reducing access to funding and tenure while increasing attrition from academia (Andalón et al., 2024).

Nobel laureates, as exemplars of scientific excellence, provide a compelling lens to probe these tensions (Betancourt et al., 2023; Dong et al., 2023; Yang et al., 2024b). Their careers challenge the lone-genius archetype, with early tendencies toward teamwork and elevated coauthorship rates pre-Prize (Li et al., 2020). Over time, their collaborative networks expand, amplifying impact. Yet, while large team sizes are well-documented among laureates (Li et al., 2019), the role of repeat collaboration in driving transformative breakthroughs—particularly Nobel-winning papers—remains underexplored. Prior studies frame repeat collaboration at the paper level, treating coauthors as a static team (Liu et al., 2022; Xu et al., 2022b, 2024; Zhang et al., 2024), but lack longitudinal, individual-level analyses linking sustained partnerships to innovation metrics like novelty and disruption.

This study fills this gap by examining repeat collaboration through dynamic ego networks, focusing on collaboration length (duration from first to last joint publication) and up-to-time strength (cumulative joint publications prior to a focal paper). We investigate how these patterns shape the innovation of Nobel laureates across Physics, Chemistry, and Medicine, offering new insights into the interplay between collaboration stability and scientific advance.

Data and method

Quantifying repeat collaboration patterns. The complexity of repeat collaboration arises from the evolving and dynamic nature of global collaboration networks. To address this, we follow the framework established by Petersen (2015). Our study adopts an ego network perspective, focusing on the career of each researcher i, with weighted links connecting the central scientist to peripheral nodes representing their collaborators j. Each career profile is constructed by aggregating publication, citation, and collaboration metadata over the first years of their career. This longitudinal approach captures author-specific factors, providing insights into how career trajectories are shaped by unique events.

Our study defines repeat collaboration as sustained coauthoring over time, characterized by two key metrics: collaboration length (the duration from the first to the last joint publication between two researchers) and up-to-time strength (the cumulative number of joint publications prior to a focal paper). This aligns closely with, yet is distinct from, the concept of persistent collaboration explored by Bu et al. (2018), who define it as the continuous, long-term coauthoring relationships that persist across multiple projects. While both concepts emphasize sustained partnerships, our focus on dynamic ego networks and the temporal evolution of collaboration (via length and strength) allows for a granular analysis of how repeated interactions influence innovation metrics.

To illustrate this concept, Fig. 1 presents the career trajectory of Thomas A. Steitz, co-recipient of the 2009 Nobel Prize in Chemistry. This schematic figure highlights two key dimensions of repeat collaborations—duration and strength:

1. Each horizontal line and its corresponding color represent a collaboration of length $l_{ij} \equiv t_{ij}^{f} - t_{ij}^{0} + 1$ between the focal scientist i and a coauthor j, beginning with their first joint publication in year t_{ij}^{0} and ending with their last observed joint publication in year t_{ij}^{f}.
2. The size of each circle denotes the total number of joint publications k_{ij} between the focal scientist i and a coauthor j.

Further analysis of Thomas A. Steitz's collaboration length distribution l reveals a power-law pattern (Fig. 1b). Additionally, we observe a strong positive correlation between collaboration length l and collaboration strength k (Fig. 1c). Our framework also enables the quantification of collaboration trajectories, with each line in Fig. 1d representing the unique trajectory of a specific collaborator pair. Our framework additionally allows us to track collaboration trajectories over time, with each line in Fig. 1d representing the distinct path of a specific collaborator pair. Using this approach, we quantify the "up-to-time" strength $k(t)$—the cumulative number of joint publications prior to a given paper, based on its publication year. This enables us to calculate other time-dependent metrics, such as up-to-time productivity and citations, which serve as control variables in our analysis.

Measures of innovation. To assess the impact and innovative nature of scientific papers, we employ four key metrics: Citation, Novelty, Disruption, and Interdisciplinarity. Each metric is adjusted to account for field-specific and temporal variations, ensuring meaningful comparisons across different disciplines and publication years.

(1) *Citation*
Citation count captures the number of citations received by scientific papers. It serves as a fundamental and widely accepted metric for gauging the academic influence and significance of research papers. However, raw citation counts present several challenges, including varying citation windows, field-specific differences, and a highly skewed distribution where a small fraction of papers receive the majority of citations. To address these issues, we transform raw citation counts into percentiles within each field and publication year. This transformation ranks papers based on their citation performance within their respective groups. In cases where multiple papers share the same citation count, we assign them the median percentile value.

(2) *Novelty*
To quantify a paper's novelty, we follow the methodology of Uzzi et al. (2013), which assesses how knowledge is recombined in scientific work. Using a citation network, we conduct ten Monte Carlo simulations to generate randomized networks by reshuffling citation links while maintaining temporal and distributional properties. This allows us to compare real-world knowledge combinations against a randomized baseline. We evaluate the novelty of a paper by analyzing journal pairings in its reference list. Each observed journal pair is compared against expected values derived from randomized networks, generating a standardized z-score:

$$Z(NoveltyU)_{i,j} = \frac{obs(pair_{ij}) - \exp(pair_{ij})}{\sigma(pair_{ij})},$$

where $obs(pair_{ij})$ is the observed frequency of a journal pair, $\exp(pair_{ij})$ is the mean expected frequency from simulations, and $\sigma(pair_{ij})$ is the standard deviation. Each paper has a unique distribution of these z-scores, representing the extent to which its references form novel knowledge combinations. Following Uzzi et al. (2013), we summarize this distribution using the 10th percentile value, denoted as $Novelty_{10p}$, which is then transformed into a percentile ranking within each field and publication year.

(3) *Disruption*
The disruption index (Funk and Owen-Smith, 2017; Park et al., 2023; Wu et al., 2019) quantifies the extent to which a paper alters the trajectory of future research. It is defined as:

$$Disruption = p_D - p_C = \frac{n_i - n_j}{n_i + n_j + n_k},$$

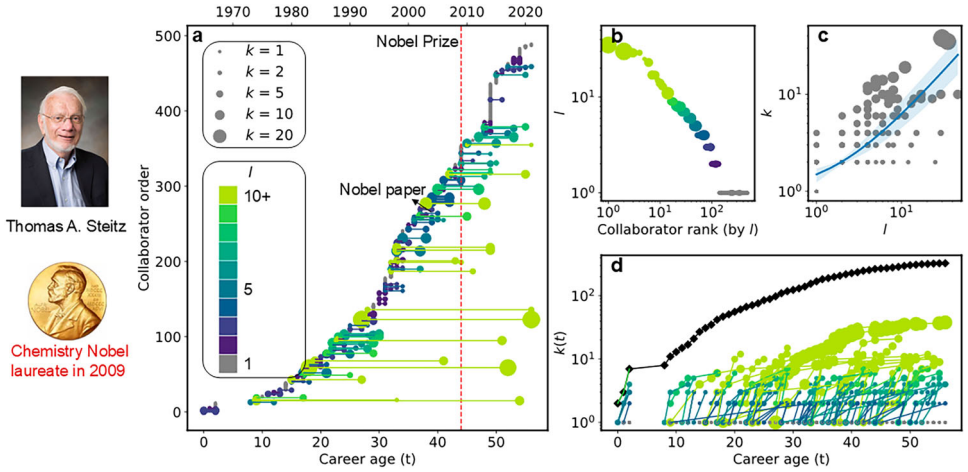


Fig. 1 Quantifying repeat collaboration in the career of Thomas A. Steitz. a Schematic representation of Thomas A. Steitz's collaboration network over time. Each horizontal line represents a collaboration between Steitz (focal scientist i) and a coauthor j, with the line length indicating the collaboration duration $l_{ij} = t_{ij}^{f} - t_{ij}^{0} + 1$, where t_{ij}^{0} is the year of the first joint publication and t_{ij}^{f} is the year of the last. Line colors distinguish unique coauthors. The size of each circle at the end of a line represents the total number of joint publications k_{ij} between Steitz and coauthor j. **b** Distribution of collaboration lengths l_{ij} across Steitz's coauthors, exhibiting a power-law pattern. **c** Scatter plot showing the positive correlation between collaboration length l_{ij} and collaboration strength k_{ij} for Steitz's coauthor relationships. **d** Collaboration trajectories over time, with each line representing the cumulative number of joint publications k(t) (up-to-time strength) between Steitz and a specificcoauthor j up to a given publication year.

where n_i represents the number of future papers citing the focal paper (FP) that do not reference its cited sources, n_j denotes the number of future papers citing FP and its references, and n_k represents the number of papers that cite the FP's references without citing the FP itself.

A high disruption index suggests that a paper introduces a paradigm shift by generating knowledge that is referenced independently of prior literature. Conversely, a low Disruption Index indicates that the paper consolidates existing knowledge, as future studies continue citing both the paper and its references. Since the disruption index is highly skewed, decreases over time, and is sensitive to citation windows, we transform raw values into percentiles within each field and publication year. This allows for meaningful comparisons, and papers with identical values are assigned the median percentile ranking.

(4) *Interdisciplinary*

To measure a paper's interdisciplinary nature, we analyze the diversity of disciplines represented in its reference list. Using the MAG dataset, which categorizes research into 292 second-level disciplines, we construct a field distribution vector for each paper based on its references. To quantify interdisciplinarity, we apply the Rao-Stirling index:

$$Rao - Stirling = \sum_{m \neq n} d_{mn} p_m p_n,$$

where p_m denotes the fraction of references in discipline m, and d_{mn} is the distance between field m and n.

For a given paper i with a reference set $R_i = \{r1, r2, \dots\}$, we map the disciplines of its references into a field distribution vector $F_i = \{field_{r1}, field_{r2}, \dots, field_{rk}, \dots\}$. The field-to-field distance d_{mn} is computed based on citation networks in MAG. Specifically, we construct a co-citation distribution vector V_m for each discipline m, reflecting its citation interactions with the other 291 fields. We then calculate the cosine distance to determine the field-to-field distance:

$$d_{mn} = 1 - CosSim(V_m, V_n).$$

As with the other metrics, raw Rao-Stirling values are converted into percentiles within each field and publication year to enable fair comparisons. Papers with identical values receive the median percentile ranking.

Fixed effect regressions. To analyze the relationship between repeat collaboration and innovation indices, we construct a regression dataset that includes more observations than unique papers. This dataset is structured as a panel by coauthor pair, meaning that papers appear multiple times if classified under multiple coauthor pairs. Ordinary least squares (OLS) regression models are employed to investigate this relationship.

To account for heterogeneity across temporal trends, research topics, and the academic status of focal scientists, we implement the following modeling strategies: (1) The regressions are conducted separately for each field to account for discipline-specific effects. (2) Year Fixed Effects (Y_t)—These control for fluctuations and trends over time by accounting for publication year differences. (3) Focal Author Fixed Effects (A_a)—These capture heterogeneity across different focal scientists (e.g., Nobel laureates), controlling for author-specific factors.

In addition to these fixed effects, I include team size as a control variable to account for team composition effects. We also incorporate the up-to-date productivity and citation count of the focal scientists to control for their career stage and academic status. The primary regression model is specified as follows:

$$ln(innovation_i) = \alpha + \beta_1 (k(t)_i) + \sum_{k=2}^{K} \beta_k \left(Control_{k,i}\right) + Y_t + A_a + \epsilon_i,$$

where i represents the article, $k(t)_i$ denotes collaboration strength up to the focal paper (i.e., the number of joint publications before the focal paper), Y_t denotes year fixed effects, A_a captures author fixed effects, ϵ_i denotes the error term, and α denotes the constant term. This approach provides a rigorous assessment of the relationship between innovation measures and repeat collaboration.

To further investigate the mechanisms driving the relationship between repeat collaboration and innovation, we introduce interaction effects between reference age and the career age difference between coauthor pairs. The modified regression

Table 1 Description of Nobel laureates in our datasets.

Fields	Variables	Obs.	Mean	STD	Min	10%	50%	90%	Max
Physics	Prize year	141	1982.3	29.8	1914	1937	1988	2018	2024
	Prize lag	141	32.0	18.3	3	13	29	54	104
	#Collaborators	141	695.6	1686.4	3	19	119	1570	11088
	Productivity	141	188.3	188.9	16	29	132	406	1154
	Total citations	141	19,023.5	26,019.9	237	1342	10,451	44,947	176,757
Chemistry	Prize year	130	1985.2	27.7	1914	1945.8	1989	2017	2023
	Prize lag	130	34.3	15.3	4	16.9	33	49.1	111
	#Collaborators	130	490.3	922.7	4	30	259.5	962.7	7952
	Productivity	130	301.4	303.2	5	37.9	190.5	652.2	1658
	Total citations	130	30,190.0	31,253.5	148	1349.2	22,773	70,716.1	145,982
Medicine	Prize year	152	1981.6	27.5	1912	1945	1984	2015.9	2022
	Prize lag	152	31.6	17.1	1	14	29	48	127
	#Collaborators	152	461.0	663.9	6	23.4	202.5	1169.7	4040
	Productivity	152	197.7	175.9	12	34.3	138.5	427.3	1085
	Total citations	152	27,006.1	26,882.7	786	3053.2	17,828.5	66,530.5	129,791

model is expressed as:

$$ln(innovation_i) = \alpha + \beta_1\left(k(t)_i\right) + \beta_2\left(age\ diff\right) + \beta_3\left(k(t)_i \times age\ diff\right) + \sum_{k=2}^{K}\beta_{k+2}\left(Control_{k,i}\right) + Y_t + A_a + \epsilon_i,$$

where: *age diff* represents the career age difference between collaborating authors, $k(t)_i \times age\ diff$ captures the interaction effect between collaboration strength and career age difference. By incorporating these interaction terms, this model aims to uncover the underlying mechanisms through which repeat collaboration influences innovation.

Publication data. Our primary dataset is derived from the Microsoft Academic Graph (MAG) (Wang et al., 2020), a comprehensive repository of scientific publications that spans from 1800 to 2021. MAG contains over 200 million documents, including journal articles, conference proceedings, preprints, and other research outputs, along with extensive bibliographic details, authorship information, author affiliations, and citation networks. For this study, we focus on publications across all types, placing particular emphasis on author career-level data. The dataset provides both dynamic data, capturing annual metrics for individual scientists, and aggregated data, reflecting long-term publication trends over the course of their careers.

To construct a robust dataset centered on author career trajectories and dynamic collaboration networks, we account for common challenges such as name disambiguation. MAG addresses this issue using a combination of advanced methodologies, including machine learning, crowdsourcing, and web-based resources such as personal websites and curricula vitae. These techniques have been validated for accuracy by Lin et al. (2023), who demonstrated that MAG's name disambiguation approach achieves 100% precision and 84% recall in distinguishing individual researchers. This ensures the reliability of our dataset in tracking career patterns and collaboration dynamics within the scientific community.

Data of Nobel laureates. For this study, we utilize an updated version of the comprehensive open-access dataset on the scientific careers of Nobel laureates, originally compiled by Li et al. (2019) and manually extended to include laureates up to 2024. The original dataset integrated unstructured information from multiple sources, including the official Nobel Prize website, personal and institutional web pages of laureates, Wikipedia entries, and publication and citation records from the Microsoft Academic Graph (MAG), which covers publications through 2021. To ensure data reliability, Li et al. (2019) employed four validation approaches: comparisons with manually collected CVs, selected Google Scholar profiles, additional affiliation records, and random manual verification of laureates.

To extend the dataset to 2024, we conducted a manual update process spanning over 24 h to incorporate Nobel laureates in Physics, Chemistry, and Physiology or Medicine awarded between 2017 and 2024. This involved collecting data from the official Nobel Prize website to identify new laureates and their associated prize-winning work. For each new laureate, we extracted biographical details, institutional affiliations, and publication records by cross-referencing the Nobel Prize website with MAG data. To maintain consistency with the original dataset, we adhered to the same validation approaches, verifying new entries against publicly available CVs, institutional profiles, and, where available, Google Scholar profiles. For publication and citation data, we relied on MAG's existing records up to 2021, as transitioning to a new bibliographic database like OpenAlex was infeasible due to differences in author name disambiguation methodologies (e.g., distinct Author IDs). The original dataset's construction by Li et al. (2019) required over 1000 h to match laureates to MAG, underscoring the complexity of replicating this process with a different database. We acknowledge the limitation of the 2021 publication cutoff and discuss its implications in Section 6. To ensure sufficient data for analysis, we filtered the dataset to include only scientists with a minimum of five publications, resulting in a final dataset of 423 Nobel laureates across Physics (123 laureates), Chemistry (115 laureates), and Physiology or Medicine (185 laureates) from 1900 to 2024.

A key finding, reinforced by our updated analysis, is that Nobel laureates exhibit a strong tendency toward collaborative research within larger teams, challenging the notion of solitary scientific breakthroughs (Singh and Fleming, 2010). The updated dataset enables the construction of dynamic ego collaboration networks, providing deeper insights into how collaboration patterns shape scientific innovation among elite scientists. Table 1 summarizes key attributes of the Nobel laureates included in our updated analysis. The dataset comprises 141 laureates in Physics, 130 in Chemistry, and 152 in Medicine, with an average Prize year falling between 1981 and 1985, spanning a range from 1912 to 2024. These laureates demonstrate exceptionally high levels of productivity, citation impact, and collaboration. Across all fields, the average total citations exceed 19,000, the average number of publications is above 188, and the average number of collaborators surpasses 460. These characteristics enable the construction of dynamic ego collaboration networks for each laureate, providing deeper insights into their collaborative trajectories.

Table 2 Comparing the innovation measures for all papers and collaboration papers.

Fields	Variables	All papers		Collaboration papers		Diff	T values	p values
		Mean	STD	Mean	STD			
Physics	#References	8.20	10.50	9.29	14.63	−1.10	−6.33	<0.001
	Team size	15.45	47.98	32.55	82.86	−17.10	−6.46	<0.001
	#Citations	124.55	117.20	130.48	148.76	−5.93	−0.09	0.93
	Disruption	0.05	0.06	0.05	0.07	0.00	2.40	<0.05
	Rao-Stirling	0.15	0.05	0.15	0.06	0.00	0.27	0.79
Chemistry	#References	15.74	12.11	17.18	19.76	−1.43	−2.73	<0.01
	Team size	4.80	9.98	16.17	64.05	−11.37	−2.99	<0.01
	#Citations	157.10	179.62	139.36	245.49	17.74	0.31	0.75
	Disruption	0.02	0.08	0.01	0.09	0.01	2.10	<0.05
	Rao-Stirling	0.18	0.06	0.18	0.06	−0.00	−0.07	0.94
Medicine	#References	17.90	13.13	17.71	14.60	0.19	−3.89	<0.001
	Team size	4.26	3.47	24.74	84.03	−20.48	−2.25	<0.05
	#Citations	159.29	147.08	183.86	285.64	−24.57	−1.16	0.24
	Disruption	0.04	0.07	0.03	0.07	0.01	3.49	<0.001
	Rao-Stirling	0.17	0.04	0.18	0.04	−0.01	−0.13	0.89

The average time lag between a laureate's first publication and the receipt of the Nobel Prize remains approximately 30 years, consistent with the original dataset.

Table 2 compares key scientific impact indicators—reference count, team size, citation count, disruption index, and Rao-Stirling interdisciplinarity score—for all papers authored by Nobel laureates and their collaborative works (papers with at least one coauthor). The table reports the mean differences (Diff) and p values to assess statistical significance. In Physics, collaborative papers have significantly more references (Diff = −1.10, $p < 0.001$), larger team sizes (Diff = −17.10, $p < 0.001$), and slightly higher disruption (Diff = 0.00, $p < 0.05$), but no significant difference in citations (Diff = −5.93, $p = 0.93$) or interdisciplinarity (Diff = 0.00, $p = 0.79$). In Chemistry, collaborative papers show more references (Diff = −1.43, $p < 0.01$), larger team sizes (Diff = −11.37, $p < 0.01$), and higher disruption (Diff = 0.01, $p < 0.05$), with no significant differences in citations (Diff = 17.74, $p = 0.75$) or interdisciplinarity (Diff = −0.00, $p = 0.94$). In Medicine, collaborative papers have fewer references (Diff = 0.19, $p < 0.001$), larger team sizes (Diff = −20.48, $p < 0.05$), and higher disruption (Diff = 0.01, $p < 0.001$), but no significant differences in citations (Diff = −24.57, $p = 0.24$) or interdisciplinarity (Diff = −0.01, $p = 0.89$). As our analysis focuses exclusively on collaboration papers, these comparisons help contextualize our findings. Collaborative papers tend to have slightly higher citation impact, though these values remain highly skewed, justifying our decision to transform all impact and innovation measures into percentiles for meaningful comparisons across different scientific domains and time periods.

Dynamic patterns of repeat collaboration

Distributions of collaboration length. Figure 2 examines the distribution of collaboration length l (the duration of repeat collaborations) across the careers of Nobel laureates. To emphasize sustained partnerships, we exclude instances where $l = 1$, meaning the collaboration lasted only a single year. Figure 2a, e, i illustrates the average l for all collaboration pairs (i.e., between the Nobel laureate and their coauthors at any career stage). While most collaborations are relatively short to moderate in duration, a subset of partnerships persists significantly longer.

To gain insight into career-level trends rather than individual collaboration pairs, we introduce the aggregated $\hat{l}$, which represents the average collaboration length for each Nobel laureate over their entire career. Since $\hat{l}$ remains fixed for each laureate, it provides a stable reference for identifying broader patterns. Figure 2b, f, j depicts the distributions of $\hat{l}$ across Physics, Chemistry, and Medicine laureates, respectively. On average, $\hat{l}$ ranges between 5 and 6 years, roughly aligning with the duration of a Ph.D. or postdoctoral period.

We further investigate the relative collaboration duration, expressed as $l/\hat{l}$, which normalizes each collaboration length by the laureate's typical collaboration span. The distribution of $l/\hat{l}$ follows a power-law pattern, as evidenced by the log-log fit. This power-law pattern suggests that while most collaborations are short-lived, a small number of partnerships endure over many years, potentially contributing to sustained scientific impact.

To understand how collaboration length changes throughout a laureate's career, we compute the average $l/\hat{l}$ at different career stages (Fig. 2d, h, l). The results indicate a clear decline over time, suggesting that Nobel laureates tend to engage in longer collaborations earlier in their careers, while later-career collaborations are generally shorter. Additionally, we overlay the average age at which laureates receive the Nobel Prize. Notably, this moment often represents a transition point: prior to receiving the prize, the relative collaboration length $l/\hat{l}$ is typically above 1, indicating a tendency toward longer-than-average partnerships. After the award, however, $l/\hat{l}$ tends to drop below 1, reflecting a shift toward shorter-term collaborations. This result indicates a tendency to engage in a broader network of collaborators rather than maintaining prior long-term partnerships after receiving the award. This shift may stem from increased recognition, new research opportunities, or changing professional priorities post-award.

Distributions of collaboration strength. Figure 3 presents the distribution of collaboration strength k (the number of repeat collaborations) throughout the careers of all Nobel laureates in three fields (Physics, Chemistry, and Medicine). To focus on meaningful repeat collaborations, we exclude cases where $k = 1$, meaning the pair of scientists collaborated only once. Figure 3a, e, i shows the average k for all collaboration pairs (i.e., between the focal Nobel laureate and their collaborators at any point in time). The majority of collaborations exhibit few or moderate repeat collaboration frequencies.

To analyze collaboration patterns at the individual laureate level rather than the collaborator pairs, we introduce the aggregated $\hat{k}$, which represents the average k for each Nobel

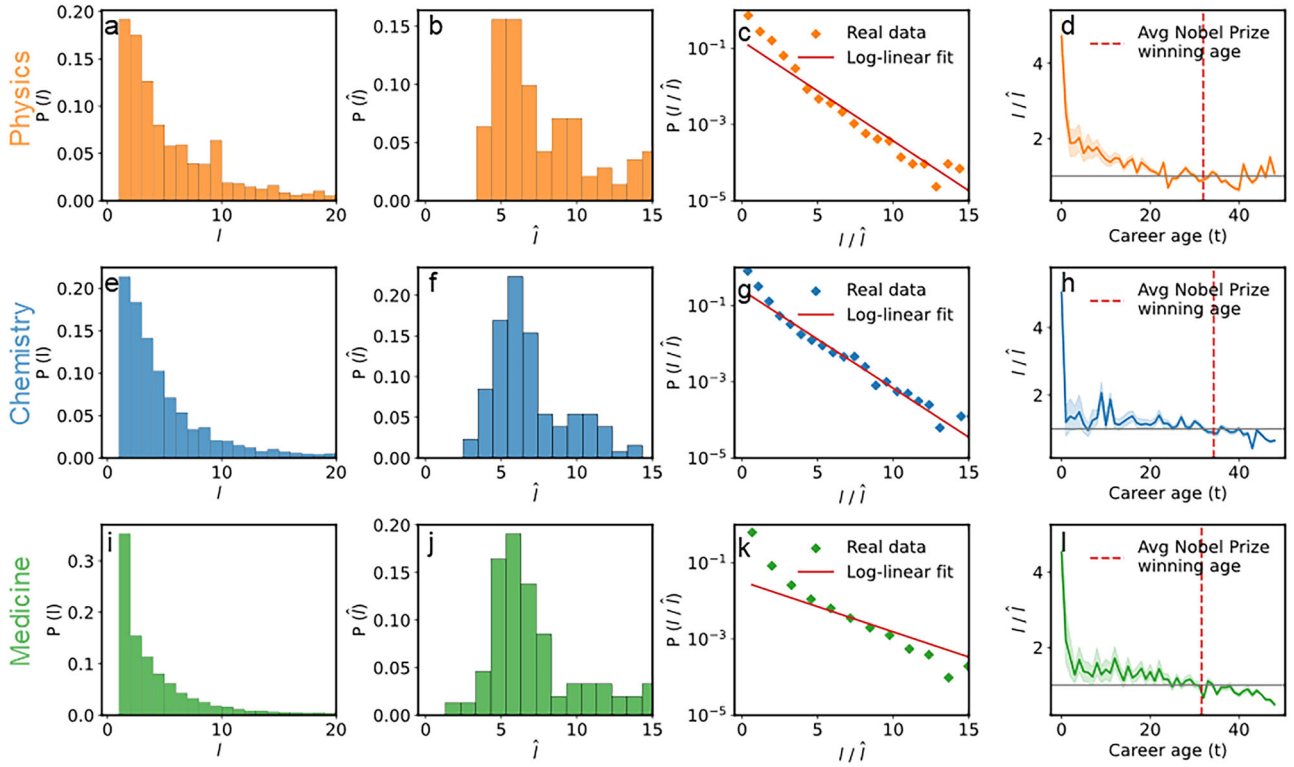


Fig. 2 Distributions of collaboration length across Nobel laureates' careers. a Distribution of collaboration length l_{ij} (duration of repeat collaborations between a Nobel laureate i and coauthor j) for Physics laureates, excluding single-year collaborations ($l_{ij} = 1$). **b** Distribution of the average collaboration length $\hat{l}$ across Physics laureates, calculated as the mean l_{ij} over each laureate's career. **c** Log-log plot of the relative collaboration length $l_{ij}/\hat{l}$, showing a power-law distribution for Physics laureates. **d** Average relative collaboration length $l_{ij}/\hat{l}$ across career stages for Physics laureates, with a vertical line indicating the average age at Nobel Prize award; values above 1 before the award reflect longer-than-averagecollaborations, while values below 1 post-award indicate shorter collaborations. **e**–**h** Same as (**a**–**d**) but for Chemistry laureates. **i**–**l** Same as **(a**–**d**) but for Medicine laureates.

laureate across their entire career. A key advantage of $\hat{k}$ is that it remains constant for each laureate, allowing us to identify stable patterns. Figure 3b, f, j reports the distributions of $\hat{k}$ for Nobel laureates in Physics, Chemistry, and Medicine, respectively. We find that the average $\hat{k}$ is between 5 and 8, indicating that most repeat collaborators work together approximately 5 to 8 times.

Next, we examine the dynamic patterns of the relative repeat collaboration rate, defined as $k/\hat{k}$, which measures the number of repeat collaborations for each collaborator pair normalized by the laureate's overall collaboration tendencies. We find that the distribution of $k/\hat{k}$ follows a power-law pattern, as shown by the log-log fit.

To understand how repeat collaboration rates evolve over a laureate's career, we calculate the average $k/\hat{k}$ at different career stages, as illustrated in Fig. 3d, h, l. The results show a consistently decreasing trend, meaning that Nobel laureates tend to collaborate more frequently with the same individuals early in their careers, but this tendency declines over time. Additionally, we plot the average age at which Nobel laureates receive their award. Notably, this age often marks a tipping point: before winning the Nobel Prize, the repeat collaboration rate $k/\hat{k}$ is typically above 1 (indicating a higher-than-average collaboration rate), whereas after winning the prize, the rate falls below 1, suggesting a decline in repeat collaborations.

Repeat collaboration pattern and Nobel-winning papers. Our primary interest lies in examining the relationship between repeat collaborations and innovation. Before conducting regression analyses on innovation measures in scientific papers, we first compare Nobel-winning papers—representing major scientific breakthroughs—with other papers authored by the same laureates. We begin by investigating whether Nobel-winning papers emerge from fresh collaborations or repeat collaborations. Since a Nobel-winning paper marks a singular moment in a laureate's career, we assess whether this pivotal achievement is more likely to result from first-time collaborations or long-standing partnerships. As shown in Fig. 4a–d, the findings suggest that in Physics and Chemistry, Nobel-winning papers are significantly less likely to be the product of repeat collaborations, meaning they are more likely to involve first-time coauthors (two-tailed t-test, $p < 0.0001$). However, in Medicine, this difference is not statistically significant, implying that Nobel-winning papers in this field are just as likely to result from new collaborations as from repeat collaborations.

Next, we compare the collaboration up-to-time strength $k(t)$—the number of joint publications before the focal paper—between Nobel-winning paper coauthor pairs and all other coauthor pairs. As illustrated in Fig. 4e–h, we find that in all three fields, Nobel-winning paper coauthor pairs exhibit significantly lower $k(t)$ values than other coauthor pairs (two-tailed t-test, $p < 0.0001$). This difference is most pronounced in Physics, where the average $k(t)$ for Nobel-winning paper coauthor pairs is only 1.9, compared to 12.4 for other coauthor pairs—a gap of approximately 6.5 times.

To further validate these findings, we control for the publication year by selecting only coauthor pairs from the same year as the Nobel-winning paper. The results, presented in Fig. 4i–l, remain consistent: in Physics and Chemistry, Nobel-winning

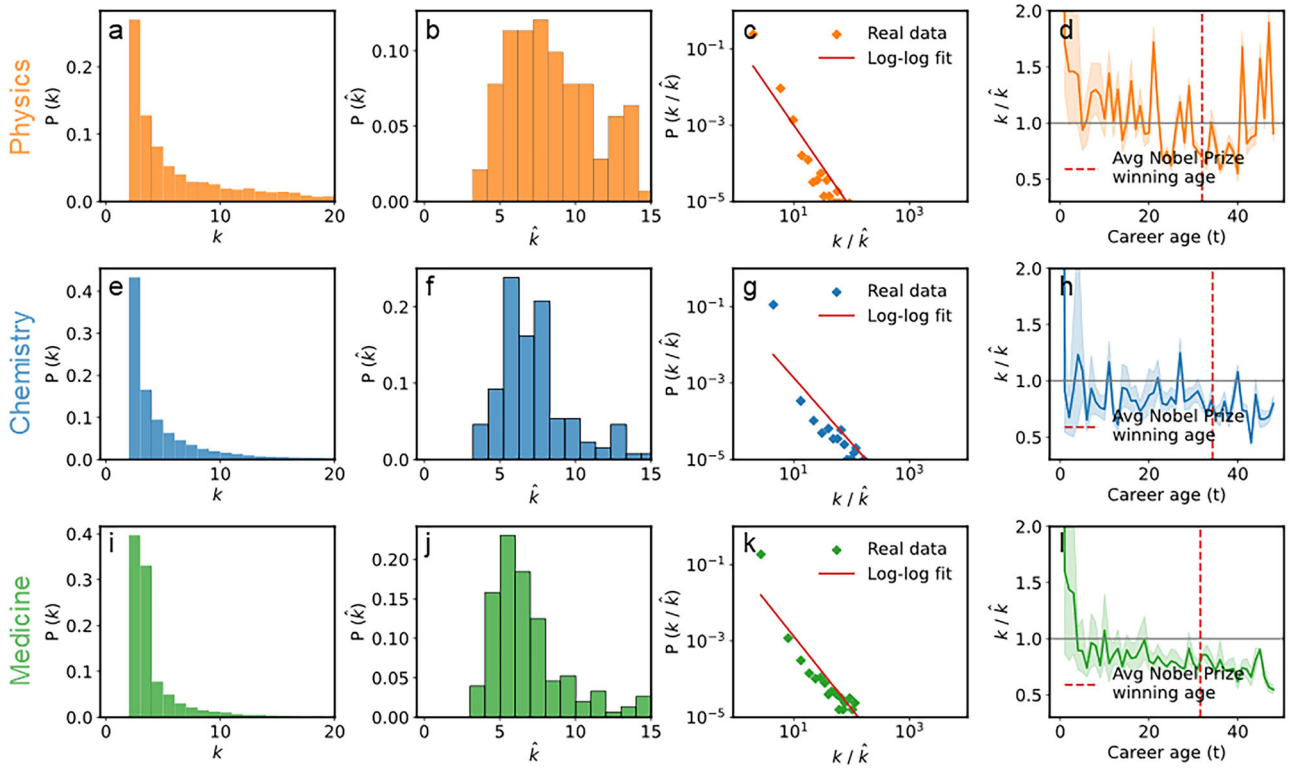


Fig. 3 Distributions of collaboration strength across Nobel laureates' careers. a Distribution of collaboration strength k_{ij} (number of repeat collaborations between a Nobel laureate i and coauthor j) for Physics laureates, excluding single collaborations ($k_{ij} = 1$). **b** Distribution of the average collaboration strength $\hat{k}$ across Physics laureates, calculated as the mean k_{ij} over each laureate's career. **c** Log-log plot of the relative collaboration strength $k_{ij}/\hat{k}$, exhibiting a power-law distribution for Physics laureates. **d** Average relative collaboration strength $k_{ij}/\hat{k}$ across career stages for Physics laureates, with a vertical line marking the average age at Nobel Prize award; values above 1 before the award indicate higher-than-average repeat collaborations, while values below 1 post-award reflect reduced repeat collaborations. **e–h** Same as (**a–d**) but for Chemistry laureates. **i–l** Same as (**a–d**) but for Medicine laureates.

paper coauthor pairs exhibit significantly lower $k(t)$ values than other coauthor pairs (two-tailed t-test, $p < 0.0001$). In Medicine, however, the difference remains insignificant.

Finally, we analyze the distribution patterns of $k(t)$ for both Nobel-winning paper coauthor pairs and all other coauthor pairs (Fig. 4m–p). We find that while other coauthor pairs follow a power-law distribution of $k(t)$, Nobel-winning paper coauthor pairs consistently display much lower $k(t)$ values. This pattern suggests that groundbreaking research is more likely to emerge from fresh collaborations rather than long-established partnerships.

Repeat collaboration and scientific innovation

Regression analysis of $k(t)$ and innovation measures. Table 3 presents the OLS regression results examining the relationship between repeat collaboration and scientific innovation across multiple fields, including Physics, Chemistry, and Medicine. In order to address the potential issues of skewed distributions and disciplinary heterogeneity, all innovation measures are transformed into percentile values. This transformation ensures that the results are comparable across different fields and publication years, making it possible to draw meaningful conclusions from the analysis.

The regression results reveal a consistent and statistically significant relationship between collaboration strength up to the focal paper $k(t)$ (the number of joint publications before the focal paper) and various innovation metrics. Specifically, the analysis shows a negative effect of $k(t)$ on the measures of citation count, disruption, novelty, and interdisciplinarity, with all p values being below 0.001. This suggests that, after controlling for other factors, an increase in the up-to-time collaboration strength is associated with a decrease in the relative innovation ranking of a paper within its field. To quantify this relationship more precisely, the results indicate that for each additional unit in $k(t)$ (i.e., an additional joint publication before the focal paper), there is a 2.67% decrease in the citation rank of the paper, a 0.27% decrease in the disruption rank, a 0.55% decrease in the novelty rank, and a 0.15% decrease in the interdisciplinarity rank. These findings suggest that as researchers engage in more repeat collaborations, their subsequent work tends to be less disruptive, less novel, and less interdisciplinary, which may reflect the tendency for highly collaborative papers to build upon established knowledge rather than introducing groundbreaking new ideas.

The regression results for different disciplines (Physics, Chemistry, and Medicine) are reported in Appendix Tables A1–A3. While the overall trend remains consistent across fields, there are some disciplinary variations: In Chemistry, the effect sizes of $k(t)$ on innovation metrics are smaller than in Physics. The relationship between $k(t)$ and interdisciplinarity (the Rao-Stirling index) in Chemistry is not statistically significant at the 0.05 level, though the direction of the effect remains negative. These variations suggest that the impact of repeat collaboration on innovation may be field-dependent, with some disciplines exhibiting stronger or weaker associations.

Mediation effect of career age difference. To further investigate the mechanisms underlying the relationship between repeat collaboration and scientific impact, we examine the mediating role of career age difference between the focal scientist (Nobel laureates)

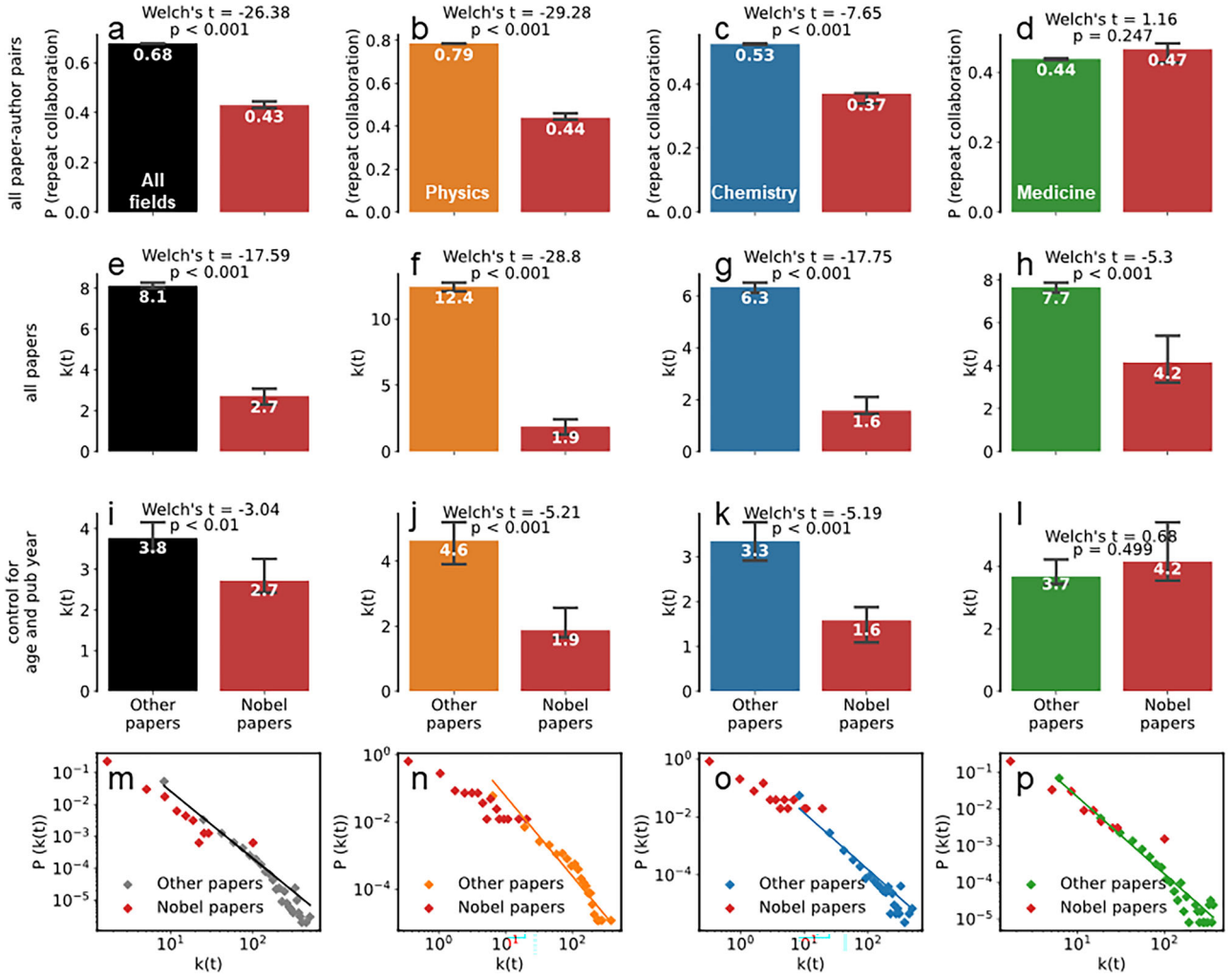


Fig. 4 Comparison of repeat collaboration patterns in Nobel-winning papers versus other papers. Proportion of Nobel-winning papers resulting from repeat collaborations compared to other papers by the same laureates, across all fields combined (**a**), Physics (**b**), Chemistry (**c**), and Medicine (**d**). Comparison of up-to-time collaboration strength k(t) (number of joint publications before the focal paper) between Nobel-winning paper coauthor pairs and other coauthor pairs, for all fields combined (**e**), Physics (**f**), Chemistry (**g**), and Medicine (**h**). **i**–**l** Same as (**e**–**h**), but controlling for publication year by comparing k(t) only for coauthor pairs from the same year as the Nobel-winning paper. Distribution of k(t) for Nobel-winning paper coauthor pairs versus other coauthor pairs, for all fields combined (**m**), Physics (**n**), Chemistry (**o**), and Medicine (**p**).

Table 3 Relationship between repeat collaboration and scientific innovation (All fields).

Models	(1)	(2)	(3)	(4)
	Citation (pct)	Disruption (pct)	Novelty (pct)	Interdisciplinarity (pct)
ln ($k(t)+1$)	−0.0267*** (0.0008)	−0.0027*** (0.0002)	−0.0055*** (0.0002)	−0.0015*** (5.62e-5)
ln (Team size)	0.3111*** (0.0018)	−0.0308*** (0.0005)	0.0106*** (0.0005)	5.31e-5 (0.0001)
ln ($t+1$)	0.8974*** (0.0102)	0.0849*** (0.0029)	0.2286*** (0.0032)	0.0463*** (0.0009)
ln (Focal author productivity + 1)	0.0576*** (0.0073)	−0.0806*** (0.0023)	−0.0261*** (0.0023)	−0.0015* (0.0007)
ln (Focal author citations + 1)	−0.3287*** (0.0046)	0.0046*** (0.0011)	−0.0520*** (0.0011)	0.0008* (0.0004)
Focal author FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	1,216,520	902,100	726,864	719,141
R^2	0.26661	0.07536	0.18041	0.21175

All models are estimated using OLS regression. Standard errors, clustered at the focal author level, are reported in parentheses.
Statistical significance is indicated as follows: *$p < 0.05$, ***$p < 0.001$.

and their coauthors. Specifically, we introduce interaction effects between reference age and career age difference within coauthor pairs to better understand how these factors influence citation impact and disruptiveness.

Figure 5a–d illustrates the mediation effect of career age difference on the relationship between repeat collaboration and citation impact, measured as citation percentiles, across all fields as well as within Physics, Chemistry, and Medicine separately.

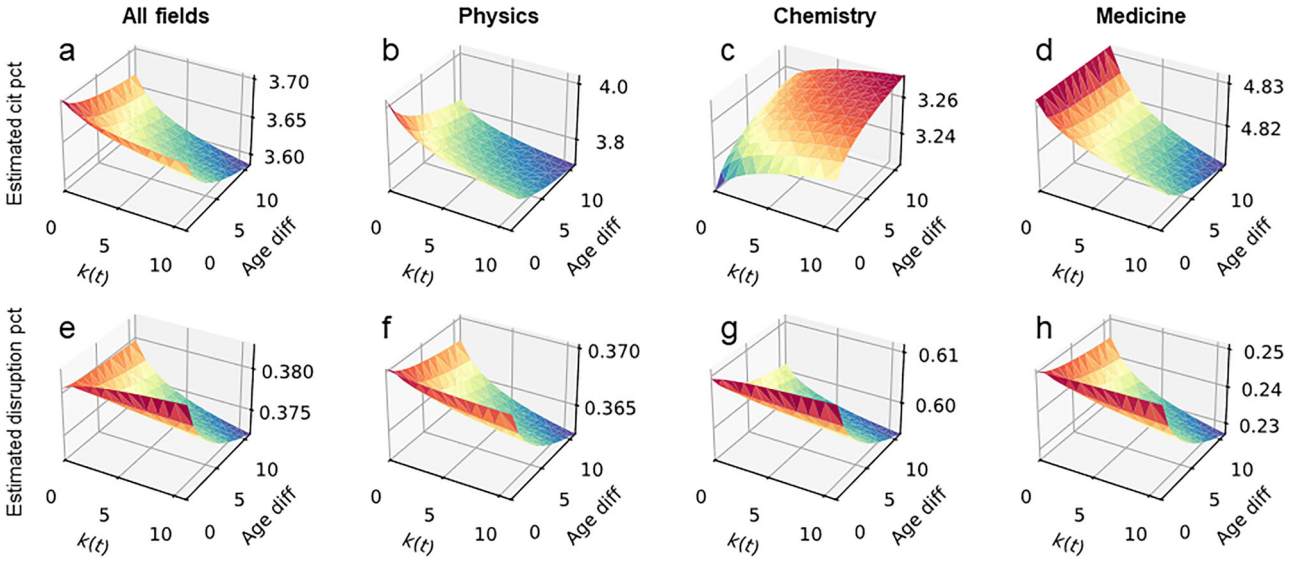


Fig. 5 Mediation effect of career age difference on repeat collaboration and scientific impact. Mediation effect of career age difference on the relationship between repeat collaboration strength k(t) and citation impact (measured as citation percentiles) for all fields combined (**a**), Physics (**b**), Chemistry (**c**), and Medicine (**d**). Mediation effect of career age difference on the relationship between repeat collaboration strength k(t) and disruptiveness (measured as disruption index percentiles) for all fields combined (**e**), Physics (**f**), Chemistry (**g**), and Medicine (**h**).

The results suggest that career age difference is negatively correlated with citation impact, indicating that larger gaps in career stages between coauthors are associated with fewer citations. Moreover, the interaction effect between reference age and career age difference on citation impact is also negative, suggesting that higher repeat collaboration strength $k(t)$ tends to be associated with larger career age gaps, which in turn lead to lower citation performance. However, this pattern does not hold in Chemistry, where career age difference exhibits a positive correlation with citation impact, implying that wider career stage gaps in Chemistry may be beneficial for citation performance.

Similarly, Fig. 5e–h reports the mediation effect of career age difference on the relationship between repeat collaboration and disruptiveness, measured as disruption index percentiles. The findings reveal that career age difference is negatively correlated with disruptiveness, suggesting that larger age gaps between coauthors are associated with less disruptive research. Furthermore, the interaction effect between reference age and career age difference on disruptiveness follows a similar negative trend, indicating that higher $k(t)$ often corresponds to larger age differences, which in turn result in research that is less disruptive.

Discussion

This study examines the relationship between repeat collaboration and scientific innovation in Physics, Chemistry, and Physiology or Medicine through the dynamic ego networks of Nobel laureates from 1900 to 2024. Using longitudinal data from the MAG and an updated Nobel laureate dataset, we quantify collaboration patterns with two metrics: collaboration length (duration from first to last joint publication) and up-to-time strength (cumulative joint publications prior to a focal paper). Innovation is assessed using four field- and time-normalized metrics: Citation, Novelty, Disruption, and Interdisciplinarity. Our findings show that Nobel-winning papers are disproportionately associated with new collaborations rather than sustained partnerships. Specifically, Nobel-winning papers in Physics and Chemistry have significantly lower up-to-time collaboration strength compared to other papers, though this difference is not significant in Medicine. Regression analyses reveal a robust negative association between repeat collaboration and innovation metrics. This effect is weaker in Chemistry, particularly for interdisciplinarity. Career age differences between coauthors mediate this relationship, with larger gaps amplifying negative impacts on citation and disruptiveness in Physics and Medicine, but showing a positive correlation with citation impact in Chemistry.

These results contribute to the Science of Science (SciSci) by quantifying how collaboration dynamics drive transformative outcomes among elite scientists. Nobel laureates, as part of the "top class" driving disproportionate impact, rely on new collaborations for breakthroughs, aligning with Granovetter's (1973) theory of weak ties, which suggests that novel information flows through transient connections. In the context of Science and Technology Studies (STS), our findings reveal how elite networks balance globalized and field-specific practices. The weaker effect in Chemistry likely reflects its reliance on cumulative, lab-based expertise, where sustained collaborations support complex experimental work, unlike the theoretical or clinical orientations of Physics and Medicine. This disciplinary variation highlights the need for tailored theoretical frameworks to account for localized scientific practices within globalized networks.

The negative association between repeat collaboration and innovation can be attributed to several mechanisms. Prolonged partnerships may lead to cognitive lock-in, where coauthors rely on familiar approaches, reducing novel knowledge recombination (reflected in lower Novelty scores). Sustained collaborations may also form socio-epistemic echo chambers, limiting exposure to diverse perspectives and constraining interdisciplinarity. In contrast, new collaborations, as weak ties, facilitate unconventional knowledge integration, driving disruption and novelty. The mediating role of career age differences adds complexity: larger experience gaps may hinder knowledge integration due to communication barriers or hierarchical dynamics, reducing citation impact and disruptiveness in Physics and Medicine. In Chemistry, however, larger career age gaps correlate with higher citation impact, possibly due to mentorship-driven knowledge transfer in lab-based settings, underscoring field-specific moderators.

These findings offer practical implications for researchers and institutions in Physics, Chemistry, and Medicine. Scientists aiming for transformative contributions may benefit from periodically engaging new collaborators to enhance novelty and

disruptiveness, countering the potential stagnation of entrenched partnerships. Institutions could support this through interdisciplinary grants or mentorship programs that encourage fresh team formations. However, Chemistry's distinct patterns suggest that strategies should be field-specific, with lab-intensive disciplines potentially favoring sustained collaborations. Funding bodies could foster innovation by valuing collaboration diversity alongside productivity, creating environments conducive to groundbreaking research in these fields.

This study has several limitations. First, its focus on Nobel laureates, while valuable for studying elite scientific achievement, limits generalizability to the broader scientific community, where collaboration dynamics may differ. Future research should test these patterns in diverse populations to assess their robustness. Second, the reliance on quantitative metrics—such as the disruption index, which is sensitive to citation windows, or novelty based on journal pairings—may overlook qualitative dimensions of originality, necessitating complementary approaches like expert evaluations. Third, the MAG dataset's cutoff at 2021 restricts analysis of recent collaboration patterns, particularly for laureates awarded after 2021. Transitioning to a database like OpenAlex was infeasible due to differences in author name disambiguation, which would require extensive rematching efforts beyond our resources. While the longitudinal depth of laureates' careers mitigates this impact, the temporal gap may affect the analysis of recent trends.

A critical limitation is the potential for endogeneity, which could arise from reverse causality or unobserved factors. For example, researchers may choose repeat collaborators for incremental projects and new collaborators for exploratory ones, influenced by project type (e.g., applied vs. fundamental research), funding constraints, or risk preferences. Similarly, a scientist's inherent ability or strategic collaboration choices may simultaneously affect collaboration patterns and innovation outcomes. The observational nature of the dataset, lacking exogenous variations (e.g., randomized collaboration assignments), makes mitigating endogeneity challenging. Advanced causal inference methods, such as instrumental variables or propensity score matching, require valid instruments (e.g., funding shocks or policy changes), which are not consistently documented here. Future studies could leverage natural experiments or qualitative data to disentangle these dynamics, though this would require expanded datasets.

Data availability

The primary dataset used in this study is derived from the Microsoft Academic Graph (MAG), which is publicly available for download at https://zenodo.org/records/6511057. The updated Nobel laureate dataset and publication career data are available at https://github.com/AlexJieYang/HSSC_repeat_collaboration.

Code availability

The analysis code used to generate the results presented in this study is publicly available on GitHub at https://github.com/AlexJieYang/HSSC_repeat_collaboration.

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References

Akbaritabar A, Castro Torres AF, Larivière V (2024) A global perspective on social stratification in science. Humanit Soc Sci Commun 11(1):914. https://doi.org/10.1057/s41599-024-03402-w

Andalón M, de Fontenay C, Ginther DK, Lim K (2024) The rise of teamwork and career prospects in academic science. Nat Biotechnol 42(8):1314–1319. https://doi.org/10.1038/s41587-024-02351-8

Betancourt N, Jochem T, Otner SMG (2023) Standing on the shoulders of giants: How star scientists influence their coauthors. Res Policy 52(1):104624. https://doi.org/10.1016/j.respol.2022.104624

Bu Y, Ding Y, Liang X, Murray DS (2018) Understanding persistent scientific collaboration. J Assoc Inf Sci Technol 69(3):438–448. https://doi.org/10.1002/asi.23966

Dong X, Lin K, Gao Y, Hu B (2023) Nobel citation effects on scientific publications: a case study in physics. Inf Process Manag 60(4):103410. https://doi.org/10.1016/j.ipm.2023.103410

Fortunato S, Bergstrom CT, Boerner K, Evans JA, Helbing D, Milojevic S, Petersen AM, Radicchi F, Sinatra R, Uzzi B, Vespignani A, Waltman L, Wang D, Barabasi A-L (2018) Science of science. Science 359(6379):eaao0185. https://doi.org/10.1126/science.aao0185

Fronczak A, Mrowinski MJ, Fronczak P (2022) Scientific success from the perspective of the strength of weak ties. Sci Rep. 12(1):5074. https://doi.org/10.1038/s41598-022-09118-8

Funk RJ, Owen-Smith J (2017) A dynamic network measure of technological change. Manag Sci 63(3):791–817. https://doi.org/10.1287/mnsc.2015.2366

Granovetter MS (1973) The strength of weak ties. Am J Sociol 78(6):1360–1380. https://doi.org/10.1086/225469

Hill R, Stein C (2025) Race to the bottom: competition and quality in science*. Q J Econ. https://doi.org/10.1093/qje/qjaf010

Johnson EJ, Bellman S, Lohse GL (2003) Cognitive lock-in and the power law of practice. J Mark 67(2):62–75

Jones BF, Wuchty S, Uzzi B (2008) Multi-university research teams: shifting impact, geography, and stratification in science. Science 322(5905):1259–1262. https://doi.org/10.1126/science.1158357

Kang D, Danziger RS, Rehman J, Evans JA (2025) Limited diffusion of scientific knowledge forecasts collapse. Nat Hum Behav 9(2):268–276. https://doi.org/10.1038/s41562-024-02041-0

Kreimer P (2022) Constructivist Paradoxes Part 1: critical thoughts about provincializing, globalizing, and localizing STS from a non-hegemonic perspective. Engag Sci Technol Soc 8(2):159–175–159–175

Leydesdorff L, Wagner CS, Bornmann L (2019) Interdisciplinarity as diversity in citation patterns among journals: Rao-Stirling diversity, relative variety, and the Gini coefficient. J Informetr 13(1):255–269. https://doi.org/10.1016/j.joi.2018.12.006

Li EY, Liao CH, Yen HR (2013) Co-authorship networks and research impact: a social capital perspective. Res Policy 42(9):1515–1530. https://doi.org/10.1016/j.respol.2013.06.012

Li J, Yin Y, Fortunato S, Wang D (2019) A dataset of publication records for Nobel laureates. Sci Data 6(1):33. https://doi.org/10.1038/s41597-019-0033-6

Li J, Yin Y, Fortunato S, Wang D (2020) Scientific elite revisited: patterns of productivity, collaboration, authorship and impact. J R Soc Interface 17(165):20200135. https://doi.org/10.1098/rsif.2020.0135

Lin Y, Frey CB, Wu L (2023) Remote collaboration fuses fewer breakthrough ideas. Nature 623(7989):987–991. https://doi.org/10.1038/s41586-023-06767-1

Liu L, Jones BF, Uzzi B, Wang D (2023) Data, measurement and empirical methods in the science of science. Nat Hum Behav 7(7):1046–1058. https://doi.org/10.1038/s41562-023-01562-4

Liu M, Jaiswal A, Bu Y, Min C, Yang S, Liu Z, Acuna D, Ding Y (2022) Team formation and team impact: the balance between team freshness and repeat collaboration. J Informetr 16(4):101337. https://doi.org/10.1016/j.joi.2022.101337

Nielsen MW, Andersen JP (2021) Global citation inequality is on the rise. Proc Natl Acad Sci USA 118(7):e2012208118. https://doi.org/10.1073/pnas.2012208118

Palla G, Barabási A-L, Vicsek T (2007) Quantifying social group evolution. Nature 446(7136):664–667. https://doi.org/10.1038/nature05670

Park M, Leahey E, Funk RJ (2023) Papers and patents are becoming less disruptive over time. Nature 613(7942):138–144. https://doi.org/10.1038/s41586-022-05543-x

Petersen AM (2015) Quantifying the impact of weak, strong, and super ties in scientific careers. Proc Natl Acad Sci USA 112(34):E4671–E4680. https://doi.org/10.1073/pnas.1501444112

Rzhetsky A, Foster JG, Foster IT, Evans JA (2015) Choosing experiments to accelerate collective discovery. Proc Natl Acad Sci USA 112(47):14569–14574. https://doi.org/10.1073/pnas.1509757112

Singh J, Fleming L (2010) Lone inventors as sources of breakthroughs: myth or reality? Manag Sci 56(1):41–56. https://doi.org/10.1287/mnsc.1090.1072

Sourati J, Evans JA (2023) Accelerating science with human-aware artificial intelligence. Nat Hum Behav 7(10):1682–1696. https://doi.org/10.1038/s41562-023-01648-z

Stirling A (2007) A general framework for analysing diversity in science, technology and society. J R Soc Interface 4(15):707–719. https://doi.org/10.1098/rsif.2007.0213

Uzzi B, Mukherjee S, Stringer M, Jones B (2013) Atypical combinations and scientific impact. Science 342(6157):468–472. https://doi.org/10.1126/science.1240474

Wang D, Song C, Barabási A-L (2013) Quantifying long-term scientific impact. Science 342(6154):127–132. https://doi.org/10.1126/science.1237825

Wang K, Shen Z, Huang C, Wu C-H, Dong Y, Kanakia A (2020) Microsoft Academic Graph: when experts are not enough. Quant Sci Stud 1(1):396–413. https://doi.org/10.1162/qss_a_00021

Wu LF, Wang DS, Evans JA (2019) Large teams develop and small teams disrupt science and technology. Nature 566(7744):378–+. https://doi.org/10.1038/s41586-019-0941-9

Wuchty S, Jones BF, Uzzi B (2007) The increasing dominance of teams in production of knowledge. Science 316(5827):1036–1039. https://doi.org/10.1126/science.1136099

Xu F, Wu L, Evans J (2022a) Flat teams drive scientific innovation. Proc Natl Acad Sci USA 119(23):e2200927119. https://doi.org/10.1073/pnas.2200927119

Xu H, Bu Y, Liu M, Zhang C, Sun M, Zhang Y, Meyer E, Salas E, Ding Y (2022b) Team power dynamics and team impact: new perspectives on scientific collaboration using career age as a proxy for team power. J Assoc Inf Sci Technol 73(10):1489–1505. https://doi.org/10.1002/asi.24653

Xu H, Liu M, Bu Y, Sun S, Zhang Y, Zhang C, Acuna DE, Gray S, Meyer E, Ding Y (2024) The impact of heterogeneous shared leadership in scientific teams. Inf Process Manag 61(1):103542. https://doi.org/10.1016/j.ipm.2023.103542

Yang AJ, Xu H, Ding Y, Liu M (2024a) Unveiling the dynamics of team age structure and its impact on scientific innovation. Scientometrics 129(10):6127–6148. https://doi.org/10.1007/s11192-024-04987-4

Yang AJ, Zhao SX, Deng S (2024b) Revisiting delayed recognition in science: a large-scale and comprehensive study. J Inf Sci. https://doi.org/10.1177/01655515241244462

Zeng A, Fan Y, Di Z, Wang Y, Havlin S (2021) Fresh teams are associated with original and multidisciplinary research. Nat Hum Behav 5(10):1314–1322. https://doi.org/10.1038/s41562-021-01084-x

Zeng A, Fan Y, Di Z, Wang Y, Havlin S (2022) Impactful scientists have higher tendency to involve collaborators in new topics. Proc Natl Acad Sci USA 119(33):e2207436119. https://doi.org/10.1073/pnas.2207436119

Zhang M-Z, Wang T-R, Lyu P-H, Chen Q-M, Li Z-X, Ngai EWT (2024) Impact of gender composition of academic teams on disruptive output. J Informetr 18(2):101520. https://doi.org/10.1016/j.joi.2024.101520

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Author contributions

AJY conceptualized and designed the study, wrote the main manuscript text, and supervised the research. JG conducted the experiments and data analysis. YS contributed to methodology development and assisted in data interpretation. SD prepared figures and contributed to manuscript revisions. SXZ provided critical feedback and helped refine the manuscript. All authors reviewed and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was not required as the study did not involve human participants.

Informed consent

This study does not contain any studies with human participants by any authors.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1057/s41599-025-05887-5.

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